Machine Learning-Based Analysis of Celestial Objects Using the SDSS17 Dataset

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***Abstract*— This study explores the application of machine learning (ML) techniques on the Sloan Digital Sky Survey (SDSS) Data Release 17 dataset, for the classification, regression and clustering of celestial objects, including stars, galaxies and quasars, using spectral features, redshift and position coordinates. Using Python and libraries such as scikit-learn, LightGBM and XGBoost, the research applies the classification algorithm (Logistic Regression, k-Nearest Neighbors, Random Forest), regression (Multiple Linear Regression, Polynomial Regression, LightGBM) and clustering (K-means, DBSCAN) on this data. The results indicate a maximum accuracy of 0.9796 in the classification with Random Forest, obtained by integrating all features, and a coefficient of determination of 0.4328 in the regression with k-NN, highlighting the combined contribution of the data. In clustering, K-means achieved a silhouette score of 0.6692 per redshift, reflecting the ability of distance-based separation. The study highlights the potential of SDSS17 as an "open science" resource for the practical application of artificial intelligence in astronomy. The conclusions confirm the effectiveness of nonlinear methods in cosmic data analysis and the essential role of open access to data in advancing research.**

Keywords*— *Machine Learning, Regression, Classification, Clustering, Stellar Classification Dataset - SDSS17

I. INTRODUCTION

The Sloan Digital Sky Survey (SDSS) is a landmark in modern astronomy, one of the largest and most influential projects to map the universe [1]. By mapping the vast expanse of space, the SDSS has created the most detailed 3D map of the universe, revealing for the first time its large-scale structure, composed of interconnected voids and filaments [2]. The SDSS data have become a central pillar in testing the standard cosmological model, demonstrating the dominance of dark matter and dark energy [3].

The success of the Sloan Digital Sky Survey (SDSS) is based on an innovative combination of tools and automated processing software systems that using public classification algorithms that combine traditional astrophysical techniques with machine learning elements [4].

The project has consistently published datasets in the form of Data Releases (from DR1 to the most recent, DR19, published in July 2025) [5]. The Sloan Digital Sky Survey (SDSS) database is structured to handle two main types of data: photometric and spectroscopic. The photometric records, which contain data for over 470 million objects, include position and brightness in five color bands (u, g, r, i, z), and are essential for determining distances and preliminary classification [6]. The spectroscopic records, although representing a much smaller subset (over 4.5 million objects) [7], are the most valuable for astrophysics, as they allow for precise classification (star, galaxy, quasar) and a measurement of redshift with superior accuracy [8].

The public access policy of the Sloan Digital Sky Survey (SDSS) has expanded the scientific impact of the project through citizen science projects [9], allowing discoveries by non-specialists. A notable example is the discovery of "Hanny's Voorwerp" in 2007 by a primary school teacher in the Netherlands [10]. This cloud of ionized gas, later confirmed by the Hubble Space Telescope, demonstrated the potential of public data in identifying rare astronomical phenomena [11].

Other major contributions of research based on the Sloan Digital Sky Survey (SDSS) include: the discovery of new quasars and galactic structures (such as the bimodal distribution of galaxies) that have influenced models of cosmic evolution [12]. SDSS data have also transformed "galactic archaeology" by providing precise measurements that reveal stellar streams in the halo of the Milky Way. These streams are the remnants of dwarf galaxies or star clusters that have been "devoured" by our galaxy, allowing astronomers to reconstruct the history of past cosmic collisions [13].

II. RELATED WORK

The "Stellar Classification Dataset - SDSS17" [14], derived from the Sloan Digital Sky Survey (SDSS) Data Release 17 (DR17), has attracted attention in numerous scientific articles and online reviews, being used for applying machine learning (ML) methods in the classification of celestial objects (stars, galaxies, quasars). Below are presented some recent studies, highlighting notable results and innovative contributions, with a focus on high accuracy and astronomical interpretation:

1) Stellar Object Classification Using Machine Learning. This article explores the classification of SDSS17 objects using deep neural networks (DNN). The results on the test set indicate an accuracy of 98% with redshift included and 97.92% with top 10 features, emphasizing the role of magnitudes and celestial position. [15].

2) Comparison of Machine Learning Algorithms and Feature Importance Analysis for Star Classification. This analysis compares ML algorithms on the SDSS17 dataset, achieving an accuracy of 97.86% with Random Forest. Redshift is the most important feature, while coordinates have minimal importance; the models perform best on the "star" class and worst on the "quasar" [16].

3) Prediction of the Stellar Class of a Star Based on Its Characteristics Using Machine Learning. The article proposes the prediction of the stellar class based on SDSS photometric data with Random Forest, achieving 76% accuracy and 3D visualizations for spatial clustering in color and luminosity space [17].

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These studies demonstrate the versatility of the SDSS datasets, providing results with high accuracy (over 97% in major cases for supervised classification) and innovative approaches, such as feature importance analysis, optimized sampling, and unsupervised clustering.

III. EXPERIMENT SETUP AND AI TOOLS

The project was developed in Python (ver. 3.10.11), using a series of standard libraries essential for artificial intelligence and data analysis. The core of the machine learning analysis was built with scikit-learn (sklearn) (ver. 1.4.2.) [18], a library that provided the necessary algorithms for classification, regression and clustering, as well as evaluation metrics and functions for cross-validation and data set partitioning. In addition, for advanced regression algorithms, the LightGBM (Light Gradient Boosting Machine, ver. 4.6.0) [19] and XGBoost (Extreme Gradient Boosting, ver. 3.0.0) [20] libraries were used, which allowed parameter optimization and superior performance. Pandas (ver. 2.1.4) [21] was used as the main tool for data preprocessing and exploration, and for data visualization, distributions and results, Matplotlib (ver. 3.7.5) [22] and Seaborn (ver. 0.13.2) [23] libraries were used. Each library had a specific and integrated role in the project workflow.

The "Stellar Classification Dataset – SDSS17" (star_classification.csv) contains 100,000 celestial objects obtained from photometric and spectroscopic measurements [18]. It includes 17 features and a target variable class, labeling each object as STAR, GALAXY, or QSO. Photometric data (u, g, r, i, z) describe object brightness in five wavelength bands, while redshift provides distance and velocity information. The celestial coordinates (alpha, delta) define the objects' positions on the sky, and additional identifiers (e.g., obj_ID, spec_obj_ID, run_ID, plate, MJD) contain observation metadata.

Redshift is a fundamental concept in astronomy describing the shift of light toward longer wavelengths as distant objects move away, caused by the expansion of space. This recession effect allows measuring both velocity and distance, forming the basis of Hubble's law [24]. Redshift values range from nearly 0 for nearby objects to higher values for distant galaxies and quasars (up to about 10–15), offering insight into the large-scale structure and evolution of the universe.

To prepare the data for the ML experiments, the following preprocessing steps were applied:

a) Initial cleaning by removing duplicates and missing values, filtering incomplete rows, and identifying anomalies and extreme values;

b) Exclusion of irrelevant features (columns), which contain information about the observation process. The following were kept: obj_ID; u, g, r, i, z (photometric data); redshift (redshift); alpha and delta; class (object class), the latter being also the target variable for classification;

c) Managing imbalance when the data presents an unbalanced distribution (approximately 59% galaxies, 22% stars, 19% quasars), corrected by adjusting the class weights in the ML models.

The study aims to demonstrate the versatility and efficiency of ML techniques by solving three fundamental astronomical problems:

a) Classification: predicting the type of object (star, galaxy, quasar) based on its characteristics.

b) Regression: estimating the value of the redshift, a key measure of the distance and velocity of objects.

c) Clustering: discovering hidden structures by grouping objects with similar characteristics.

By implementing and comparing specific algorithms, this work will analyze the ability of these models to extract valuable information from astronomical data, paving the way for new approaches in cosmological research.

IV. RESULTS

A. Classification

For Classification we have the categorical variable "class", with the values "GALAXY", "STAR" and "QSO", which we will use as the target variable.

Logistic Regression; k-Nearest Neighbors (k-NN); Random Forests [18] and XGBoost [20] algorithms were applied to classify celestial objects according to the spectral characteristics u, g, r, i, z (ultraviolet, green, red, near infrared, infrared).

The metrics of accuracy (accuracy - proportion of correct predictions), precision (precision - proportion of correctly identified positive cases), recall (sensitivity - proportion of real positive cases that the model managed to detect) and F1-score (F1-score - a balanced average between precision and sensitivity) were followed.

Stratification, a sampling technique that divides the data set into subgroups, was used, ensuring that the proportion of each class (e.g., galaxy, qso, star) is represented equally in each subgroup, to obtain a distribution similar to that of the original data.

The k-NN and Random Forest algorithms [18] were applied, for the spectral features (u, g, r, i, z,), position ("alpha" and "delta") and redshift, respectively, on all these features (Table 1, Fig. 1).

Random Forest demonstrated the highest accuracy (0.9796) when applied to the full feature set, slightly outperforming XGBoost (0.8686) and significantly outperforming linear models such as Logistic Regression. This result suggests the superiority of nonlinear algorithms in capturing complex relationships in spectral and position data. The inclusion of the redshift feature proved particularly influential, with k-NN achieving an accuracy of 0.9492, due to the fact that redshift indicates the distance of objects through Hubble's law, which relates redshift to the rate of retreat caused by the expansion of the universe.

These results are similar to those published in other peer-reviewed papers using deep neural networks (DNNs) [15] (which show an accuracy of 98% with redshift included as a dominant feature and 97.92% with all features), and slightly superior to papers using the same algorithms [16] (which show an accuracy of 97.86% with Random Forest). The superior performance with the combined features highlights the synergistic effect of spectral, redshift and position data, with redshift acting as a dominant classifier due to the distinct distances of stars, galaxies and quasars.

Features	spectral features (u, g, r, i, z)				redshift		alpha, delta		all features	
Method/ Results	Logistic regression	k-NN	Random Forest	XGBoost	k-NN	Random Forest	k-NN	Random Forest	k-NN	Random Forest
Accuracy	0.7540	0.8592	0.8742	0.8686	0.9492	0.9244	0.6667	0.6749	0.8327	**0.9796**
precision GALAXY	0.79	0.90	0.91	0.90	0.94	0.94	0.73	0.76	0.83	**0.98**
precision QSO	0.66	0.80	0.82	0.81	0.92	0.81	0.42	0.43	0.87	**0.97**
precision STAR	0.68	0.79	0.82	0.81	1.00	1.00	0.59	0.54	0.79	**0.99**

Table 1. Results obtained by applying classification algorithms

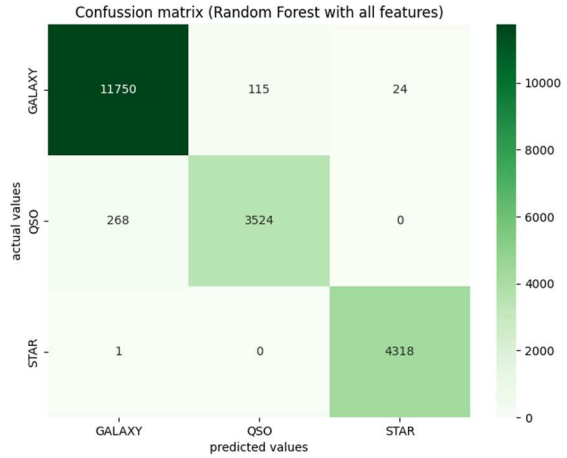


Fig. 1. Confusion matrix

A scientific explanation for the performance of the redshift feature, which makes it an exceptional feature for classification, would be that the three types of objects (galaxies, quasars and stars) are at fundamentally different distances: stars are in our galaxy or very close, having a redshift close to zero; galaxies are much more distant, having a higher redshift, and quasars are extremely luminous, but also extremely distant, having the highest redshift, therefore the value of the redshift acts as a direct indicator of the type of object, transforming a complex classification problem into a much simpler one.

The position features (alpha, delta) showed low discriminative power, due to their 2D projection nature, where the positions on the sky of different types of objects overlap, a star in the Milky Way may appear on the sky in the same direction as a distant galaxy or a quasar.

If the model had analyzed larger data sets, it would have observed that most stars cluster along the galactic plane, forming a dense band, while galaxies and quasars have a more uniform distribution on the sky, away from the galactic plane. Therefore, position is not an individual classification factor, but an element of a more complex data structure. But even these characteristics, together with the spectral ones and especially the redshift, led to remarkable results.

B. Regression

For Regression, the redshift feature was chosen, as a continuous measure was the most appropriate.

Multiple Linear Regression, Polynomial Regression of degrees 2, 3 and 4 were applied for comparative analysis; k-Nearest Neighbors (k-NN) for regression, where the performance of the algorithm was evaluated based on neighbors; LightGBM, where various regularization parameters were adjusted to control the complexity of the model [19]. Spectral and position features were used, separately and together.

In regression analysis, the performance of the models was evaluated through metrics such as the Coefficient of Determination (R2), which indicates the proportion of the variation in the dependent variable that can be explained by the independent variables, and the mean squared error (MSE), which measures the average of the squared errors between the predicted and actual values. The training and testing sets were balanced using the stratification technique, respectively the cross-validation technique.

For Linear Regression, Polynomial Regression and LightGBM the best results were obtained on all features (spectral and position) and for k-NN the best results were obtained only for spectral features.

Method /Metrics	Linear regression	Polynomial regression of degree 2	k-NN (k=22)	LightGBM
R^2	0.2792	0.2939	**0.4328**	0.4259
MSE	0.3902	0.3823	0.3071	0.3070

Table 2. Results obtained by applying regression algorithms

If we make a comparison of the methods used and the results obtained, it can be seen that the k-NN algorithm achieved the best performance among all the methods tested, applied only to spectral features, closely followed by LightGBM (Table 2).

For the k-NN method for regression, the parameter k – the number of neighbors – was adjusted to obtain the best performance, resulting in a value of k=22 (Fig. 2).

The best parameters for LightGBM were identified as: n_estimators=600 (represents the number of decision trees the model builds); learning_rate=0.015 (controls how much each tree contributes to the final prediction); max_depth=12 (sets the maximum depth of each tree); num_leaves=70 (represents the maximum number of leaf nodes in a tree); min_child_samples=30 (defines the minimum number of examples required in a node to create a new branch - a new leaf); subsample=0.9 (controls the percentage of samples from the training set that are used to build each tree); colsample_bytree=0.8 (represents the percentage of features that are randomly selected to build each tree); reg_alpha=0.1 and reg_lambda=0.1 (are regularization parameters, they add a penalty to the cost function of the model, a higher value of these parameters reduces the complexity of the model, preventing over-fitting by penalizing large branch weights) [19].

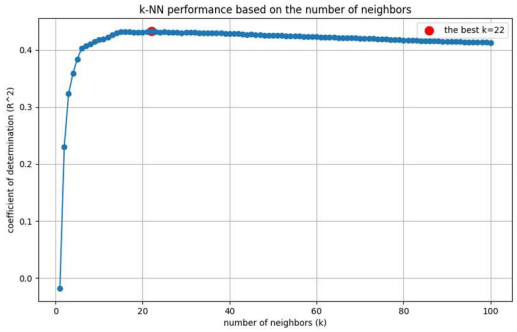


Fig. 2. Performance of the k-NN algorithm

Increasing the number of trees can enhance performance but also raises the risk of overfitting and longer training times. A lower learning rate improves robustness yet requires more trees. Deeper trees and a higher number of leaves capture complex patterns but increase model complexity. In LightGBM, which grows trees leaf-wise, these parameters are critical. Using 90% of the data per tree (subsample = 0.9) helps limit overfitting and improve diversity [19].

Linear regression performed the worst, suggesting that a simple linear relationship is not adequate to model the data. Polynomial regression of degree 2 significantly improved performance over linear regression, indicating the presence of nonlinear relationships in the data.

The results obtained demonstrate that the relationship between redshift and spectral/positional features is a non-linear one. Redshift is a complex measure, although at a simplistically level, redshift increases with distance (according to Hubble's Law [24]), the relationship is not perfectly linear on a large cosmological scale. Factors such as gravity (which can warp space-time) and the proper motions of galaxies with respect to the general flow of expansion (known as "peculiar motion" [25]) introduce non-linearities.

Distance and tree-ensemble-based algorithms, k-NN, but especially LightGBM have demonstrated a superior ability to model data compared to linear models.

C. *Clustering*

For clustering, unsupervised learning algorithms (a category of machine learning algorithms that analyze unlabeled data sets with the aim of discovering hidden structures, patterns and relationships in the data) K-means and DBSCAN (Density-Based Spatial Clustering of Applications with Noise) [18] were used, which segmented the data into clusters based on the similarity of spectral, position and redshift features, with the aim of discovering hidden structures.

Here we will use the silhouette score metric (Table 3), which measures how well an object is separated from neighboring clusters. Its values range from -1 to +1. A score of +1 indicates that the object is perfectly grouped in its cluster, while a score of -1 suggests that it has been incorrectly assigned to a cluster. A value of 0 indicates an overlap between clusters [18].

For more features we will use PCA – Principal Component Analysis, which is a dimensionality reduction technique used to simplify a data set by transforming many variables (features) into a smaller set of new variables, called principal components, with the aim of preserving as much information from the original data as possible, but in a much smaller number of dimensions.

K-means algorithms have been modeled after several values for the number of clusters (k=3, 4 and 5). The K-means algorithm is an iterative process that, by constantly calculating the means, moves the centers of the clusters to ensure that each data point is in the most appropriate group. For k=3 we check if the resulting clusters correspond to a certain extent with the three main classes (GALAXY, QSO, STAR).

In conclusion, K-means on spectral features was able to create clusters that have a tendency to group together objects of the same type, especially galaxies. However, the clusters do not perfectly match the true classes, with a significant mixture of different types of objects in each cluster (Fig. 3, Fig. 4, Fig. 5). For example, galaxies, stars and quasars do not have distinct spectral features but overlap, but redshift is more successful in separating them (the best value of the silhouette score metric being obtained for k=5). This suggests that although there are spectral similarities between objects of the same type, there are also variations or overlaps that make a clear separation difficult.

K-means achieved moderate success with redshift (silhouette score 0.6692 at k=5), reflecting its ability to separate objects based on distance, although spectral overlaps limited perfect alignment with the true classes.

DBSCAN is a good alternative because it is a density-based clustering algorithm and does not require specifying the number of clusters in advance.

Here we have two parameters to adjust, min_samples, which represents the minimum number of points in a neighborhood to form a cluster, and eps (epsilon) represents the maximum distance between two points for them to be considered in the same neighborhood.

Performance was evaluated using three metrics: homogeneity, which assesses whether each cluster contains data points from a single true class; completeness, which verifies whether all members of a class are assigned to the same cluster; and the v-measure, the harmonic mean of the two, providing a balanced indicator that penalizes both mixed and fragmented clusters [19].

Features	spectral (u, g, r, i, z)			celestial (alpha and delta)			redshift			all features		
k	3	4	5	3	4	5	4	5	6	3	4	5
silhouette score	0.3765	0.3842	0.3562	0.4727	0.4852	0.4248	0.6569	**0.6692**	0.6569	0.3015	0.2820	0.2784

Table 3. Results obtained by applying clustering algorithms

To choose eps (epsilon) for DBSCAN, a distance graph called the k-nn graph is constructed (Fig. 6). It helps us identify an optimal value for eps by identifying a "knob" on the curve of the graph. In our case, the "knob" of the curve is at approximately 0.38. This would be a good starting value for eps in the DBSCAN algorithm.

Although it was able to identify the three classes, they were not grouped properly. Algorithm a considered 811 points as noise and failed to create distinct clusters for each class (Fig. 7).

A homogeneity score of 0.00; completeness score of 0.06 and a v-measure score of 0.00 were obtained, which shows that DBSCAN completely failed to group the data correctly.

For the redshift feature, from the analysis of the k-nn distance graph, a value of 0.1 was obtained for eps, which was able to identify two classes, and the results were mediocre (homogeneity score: 0.00; completeness score: 0.18; v-measure score: 0.00).

For all features, the optimal value of eps was chosen as 1.25, a number of 4 classes were identified, but the results were not satisfactory (number of points considered as noise: 270; homogeneity score: 0.00; completeness score: 0.08 and v-measure score: 0.00).

The failure of DBSCAN to form distinct clusters, highlighted by the low v-measure scores, suggests the inadequacy of density-based methods for the feature distribution of this dataset. The analysis indicates that redshift improves clustering by distance-based separation, while position and spectral features introduce noise due to overlaps.

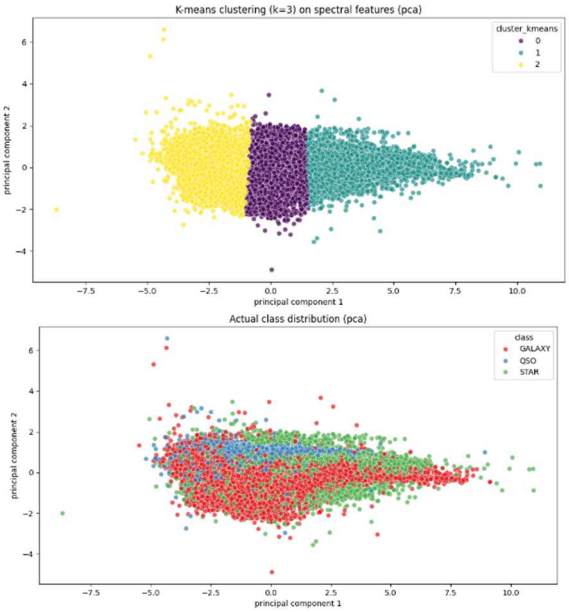


Fig. 3. K-means clustering on spectral features (for k=3)

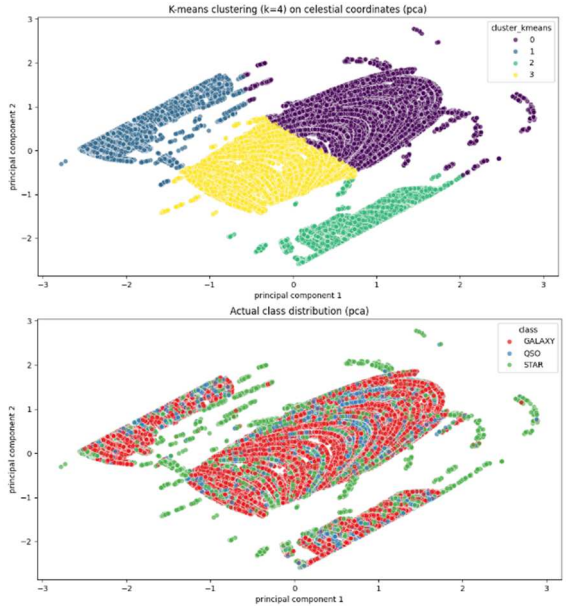


Fig. 4. K-means clustering on position features (for k=4)

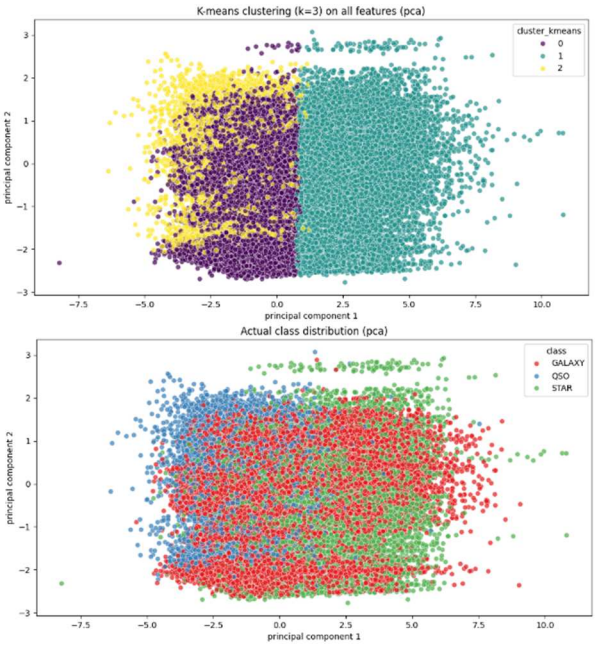


Fig. 5. K-means clustering on all features (for k=3)

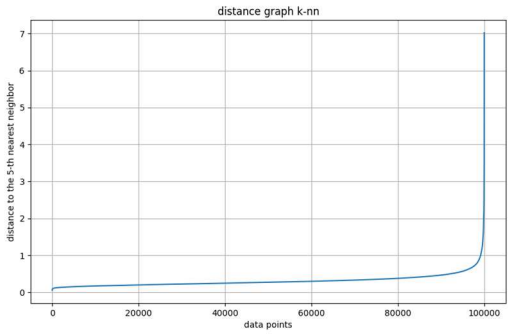


Fig. 6. Choosing the characteristics of the DBSCAN algorithm

The reported metrics represent single-run results for each algorithm. Future work will include multiple runs with different random seeds to compute standard deviations and confidence intervals, providing a more robust statistical evaluation of model performance.

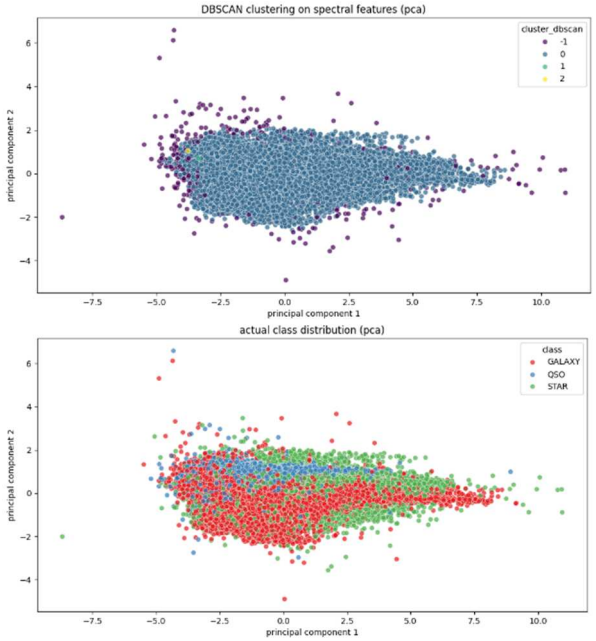


Fig. 7. DBSCAN clustering on spectral features

V. DISCUSSION AND CONCLUSIONS

This paper demonstrated the effectiveness of machine learning techniques (classification, regression, clustering) applied to the SDSS17 dataset. The remarkable performance, highlighted by the accuracy of 0.9796 obtained with Random Forest, highlights the effectiveness of nonlinear algorithms in the analysis of complex astronomical data and the determining role of the redshift feature. In addition, the study highlights the didactic value of the dataset, providing a practical platform for students and enthusiasts.

The adopted multidisciplinary approach confirms that the SDSS17 dataset, characterized by its well-organized structure and predefined labels, functions as an ideal virtual laboratory. It allows the full exploration of the life cycle of a data science project, from data preprocessing and the application of classification algorithms (e.g. k-NN with an accuracy of 0.9492 per redshift) and regression (k-NN with R^2 of 0.4328), to clustering techniques (K-means with a silhouette score of 0.6692 per redshift). This flexibility facilitates not only individual learning of the algorithms, but also the development of skills in interpreting the results, essential for future specialists.

An important educational aspect of using SDSS17 is its accessibility, enabling even non-specialists to contribute to discoveries, as demonstrated by the "Hanny's Voorwerp" case through citizen science. This openness fosters engagement among students and enthusiasts. The variability of regression and clustering outcomes (e.g., DBSCAN's low homogeneity of 0.00) further highlights the importance of selecting appropriate methods and data for reliable analysis.

In conclusion, this study confirms the efficiency of machine learning algorithms on the SDSS17 dataset and underscores the importance of "open science" in democratizing access to high-quality astronomical data. By translating theory into practice, it supports the training of a new generation of researchers able to apply artificial intelligence to fundamental scientific problems.

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